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United States Patent	12410592
Kind Code	B2
Date of Patent	September 09, 2025
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Water providing station

Abstract

A water providing station having a housing that supports a basin. A water dispensing mechanism is associated with the basin and is coupled to a water supply line located within the housing. A filter is also coupled to the water supply line. Mounted to a portion of the housing is a door that is moveable between open and closed positions. The filter is mounted to and supported by the door for movement moved with the door between the open and closed positions.

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Appl. No.:	16/941043
Filed:	July 28, 2020

Prior Publication Data

Document Identifier	Publication Date
US 20220034073 A1	Feb. 03, 2022

Publication Classification

Int. Cl.: E03B9/20 (20060101); B01D29/13 (20060101); B01D35/30 (20060101); C02F1/00 (20230101); C02F101/12 (20060101); C02F101/20 (20060101); C02F101/30 (20060101)

U.S. Cl.:

CPC E03B9/20 (20130101); B01D29/13 (20130101); B01D35/306 (20130101); C02F1/001 (20130101); B01D2201/04 (20130101); B01D2201/4092 (20130101); C02F2101/12

Field of Classification Search

CPC: E03B (9/20); E03B (7/07); B01D (29/13); B01D (35/306); B01D (2201/04); B01D (2201/4092); B01D (27/08); B01D (2201/0423); B01D (2201/303); C02F (1/001); C02F (2101/12); C02F (2101/20); C02F (2101/306); C02F (2201/006); C02F (2307/10); C02F (1/003)

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Background/Summary

BACKGROUND

Field of the Invention

(1) The present invention generally relates to water providing stations, such as drinking fountains, water bottle filling stations and hand washing stations. More specifically, the invention relates to a water providing station for use in providing filtered water for drinking, bottle filling, hand washing or other purposes.

Description of Related Art

(2) Water providing stations, such as but not limited to, drinking fountains, water bottle filling stations, hand washing stations, etc., are all coupled to water supplies to provide clean and safe tap water to the user. While such water supplies, typically a municipal water supply, generally provide good and safe water, the quality of the water may and can be improved.

(3) To improve the quality of the water, a variety of filters exist that can be incorporated into a water supply station. The construction and materials of the filter will vary depending on the particular contaminant(s) that are intended to be removed from the water. Common contaminants that are often of concern include, without limitation, chlorine, lead, mercury, pesticides and

herbicides.

(4) One common type of filter is the screw-on filter cartridge. The filter is installed and removed from the water supply station by incorporating a filter head into the water supply line. The cartridge is provided with integral fittings and a threaded connector thereby allowing the cartridge to be mounted to a corresponding threaded connector provided on the filter head. In this way, the filter can be easily removed from the water supply once its service life is complete and replaced with a new filter.

(5) The replacement of the filter still requires the removal of an access panel from the water providing station to expose an access opening into the station and to expose the filter. The service technician also must reach through the opening to access the filter, manipulate the filter to release it from the filter head, and retract the filter through the opening. The new filter is similarly installed in a reverse ordered process until the access panel cover is repositioned and secured over the access opening.

SUMMARY

(6) In overcoming the various drawbacks and other limitations of the related art, the present invention provides a water providing station in which access to a replaceable filter is made easier and more efficient. This is achieved, in one aspect of the invention, by the filter being mounted to a drop-down door and moveable with the door to an easy access position.

(7) Accordingly, in one aspect, the invention provides a water providing station having a housing that supports a basin. A water dispensing mechanism is associated with the basin and is coupled to a water supply line located within the housing. A filter is also coupled to the water supply line. Mounted to a portion of the housing is a door that is moveable between open and closed positions. The filter is mounted to and supported by the door for movement moved with the door between the open and closed positions.

(8) In another aspect, the door is pivotally mounted to the housing.

(9) In a further aspect, one end of the door is pivotally engaged with the housing and at another portion of the door the door is releasably engaged with the housing.

(10) In an additional aspect, a filter hub, coupled to the water supply line, is engaged with the filter and defines a fluid connection between the filter and the water supply line.

(11) In still another aspect, the filter hub is supported by the door.

(12) In yet a further aspect, the filter hub is fixedly fastened to the door for movement with the door between the open and closed positions.

(13) In an additional aspect, the filter hub is in threaded engagement with the filter.

(14) In yet another aspect, the door is pivotally engaged adjacent to one end of the door.

(15) In still a further aspect, the door includes a main panel defining an exterior surface, the exterior surface corresponding with an exterior surface of the housing in the closed position.

(16) In an additional aspect, the exterior surface is flush with the exterior surface of the housing in the closed position.

(17) In another aspect, the door includes a mounting bracket, the filter being connected to the mounting bracket.

(18) In a further aspect, the filter is indirectly connected to the mounting bracket.

(19) In yet an additional aspect, the mounting bracket extends away from the main panel on a side of the main panel opposite from the exterior surface of the main panel.

(20) In another aspect, a filter hub, the filter hub being mounted to the mounting bracket.

(21) In yet a further aspect, the filter hub is connected to the filter and indirectly connects the filter to the mounting bracket.

(22) In still an additional aspect, the door includes lateral flanges adjacent to at least one end of the door, the door being mounted to the housing by pivot pins extending through the lateral flanges.

(23) In yet another aspect, the filter includes coupling port and a body portion, the body portion being supported by the door.

- (24) In a further aspect, the door includes a main panel having support shelves provided along at least portions of lateral sides of the main panel.
- (25) In an additional aspect, the filter engages the support shelves.
- (26) In still another aspect, the body of the filter engages the support shelves.
- (27) Further objects, features and advantages of this invention will become readily apparent to persons skilled in the art after review of the following description, including the claims, and with reference to the drawings that are appended to and form a part of this specification.
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Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a perspective view of a incorporating the principles of the present invention.
- (2) FIG. 2 is a perspective view of the water providing station seen in FIG. 1, with portions broken away and shown in phantom to reveal a filter located within the water providing station.
- (3) FIG. 3 is perspective view of the underside of the water providing station and illustrates the hinged filter door in a closed position.
- (4) FIG. 4A is view similar to FIG. 3, with the filter door in an open position.
- (5) FIG. 4B is view similar to FIG. 4B, with the filter removed from the open filter door.
- (6) FIG. 5 is a perspective view of the filter and filter door isolated from water providing station.
- (7) FIG. 6 is an exploded perspective view of the filter and filter door seen in FIG. 5.

DETAILED DESCRIPTION

(8) As used in the description that follows, directional terms such as “upper” and “lower” are used with reference to the orientation of the elements as presented in the figures. Accordingly, “upper” indicates a direction toward the top of the figure and “lower” indicates a direction toward the bottom of the figure. The terms “left” and “right” are similarly interpreted. The terms “inward” or “inner” and “outward” or “outer” indicate a direction that is generally toward or away from a central axis of the referred to part, whether or not such an axis is designated in the figures. An axial surface is therefore one that faces in the axial direction, a direction along the central axis. A radial surface therefore faces radially, generally away from or toward the central axis. It will be understood, however, that in actual implementation, the directional references used herein may not necessarily correspond with the installation and orientation of the corresponding components or device.

(9) Referring now to the drawings, a water providing station, hereafter “water station,” embodying the principles of the present invention is generally illustrated in FIGS. 1-3 and designated at **10**. As seen in the figures, the water station **10** is shown as a drinking fountain. This is for illustration purposes only and is not intended to limit the scope or application of the principles of the present invention. As will be appreciated, the principals of the present are equally applicable to other types of water stations, including, without limitation, water bottle filling stations, glass filling stations, hand washing stations, etc.

(10) As seen in the figures, the water station **10** includes a bubbler **12** that provides water on demand for drinking purposes. The bubbler **12** is typical on drinking fountains, but would be omitted and replaced with a faucet or other water dispenser in other types of water stations. The on demand aspect of the water station **10** is achieved via a pushbutton and valve assembly **14** connected to a water supply line **15** (seen in FIG. 2) that is further coupled via an inlet connector (not shown) to a potable water source, such as a municipal water source (also not shown). The bubbler **12** provides the water upward and out over a basin **16**, which collects the excess water and directs it to a drain opening **18** in the basin **16**. Water from the drain opening **18** is collected in drain line (not shown) that may include a tailpipe coupled through a P-trap coupled to a waste pipe extending from a wall to which the water station **10** is mounted. Alternatively, the water station **10**

may be free standing and both the water source and waste pipe may emerge from the ground or an alternate location **12**. The various plumbing components of the water station **10** are enclosed in a housing **19**, generally comprised of an apron **20** and a cover **22**, upon which the basin **16** is supported.

(11) As seen in FIG. 2, the water station **10** includes an internally located filter **24**. The filter **24** is provided in-line with the water supply line **15** upstream of the bubbler **12** or other water dispenser. Preferably, the filter **24** is a screw-on, cartridge-type filter in which both the inlet and outlet of the filter **24** are located centrally of a threaded coupling **26** on one end of a body **27** of the filter **24**. Alternatively, the filter **24** may be provided in-line with the water supply line **15** through compression fittings, barbed fittings, clamp fittings, quick connect and other similar and/or conventional fittings.

(12) The filter **24** is connected to the water supply line **15** through a filter hub **28** that includes a threaded coupling **30** that matingly engages the threaded coupling **26** of the filter **24**. The filter hub **28** also includes inlet and outlet ports **32, 34** that couple with the inlet and outlet ports of the filter **24** during engagement of the threaded couplings **26, 30** and provide for fluid communication between the filter hub **28** and the filter **24**. Because of the threaded couplings **26, 30**, the filter **24** is readily disconnected from the water supply line **15** after its useful life and replaced with another filter **24**.

(13) However and as previously mentioned, access to a filter in prior water stations was cumbersome and inconvenient. In accordance with the principles of the present invention, access to the filter **24** is easily gained and removal and replacement of the filter **24** is readily achieved.

(14) Referring now to FIG. 3, provided in a bottom wall **36** of the apron **20** is a door **38**. The door **38** is hinged at one end and releasably engaged at the other end with the bottom wall **36**. As seen in FIG. 3, the door **38** is in the closed position and a locking mechanism **40**, such as a simple lock set, secures a distal end **42** of the door **38** with the bottom wall **36**. While not readily seen in FIG. 3, the proximal end **44** of the door **38** is pivotally connected with the bottom wall **36** so as to allow the distal end **42** of the door **38** to swing downward into an open position when the locking mechanism **40** is unlocked. The open position of the door **38** is seen in FIGS. 4A and 4B.

(15) As seen in FIG. 4A, the filter **24** is carried by the door **38** and is pivoted downward, into an easily accessible position, when the door **38** is moved into its open position. To facilitate this movement, the filter hub **28** is mounted to the proximal end **44** of the door **38**, also for movement with the door **38**. To remove the filter **24**, the filter **24** is rotated counter-clockwise, as indicated by arrow **46** in FIG. 4B, thereby unthreading the threaded coupling **26** of the filter **24** from the threaded coupling **30** of the filter hub **28**. To install a replacement filter **24**, the process is simply reversed and the filter **24** is rotated clockwise, in a direction opposite to arrow **46**, thereby threading the threaded coupling **26** of the filter **24** into engagement with the threaded coupling **30** of the filter hub **28**.

(16) Referring now to FIGS. 5 and 6, the filter **24**, filter hub **28** and door **38** are shown removed from the remainder of the water station **10**. As seen therein, the door **38** includes a main panel **48** that defines a surface that is flush with the bottom wall **36** of the apron **20** when the door **38** is in the closed position. At the lateral (lengthwise) sides of the door **38**, upright flanges **50** are provided and, at the proximal end **44**, the flanges **50** include apertures **52** through which extend pivot axles **54**. The pivot axles **54** may be formed by pins extended through the apertures **52** or by other mechanical fasteners, such as partially or fully threaded bolts and retaining nuts. The pivot axles **54** extend laterally outward from the flanges **50** to engage corresponding brackets on the upper (inner) side of the bottom wall **36** to form the pivot engagement between the door **38** and the bottom wall **36** of the apron **20**.

(17) Also at the proximal end of the door **38**, a flange on the end of the main panel **48** defines a mounting bracket **56** for securing the filter hub **28** to the door **38**. The mounting bracket **56** is preferably oriented 90 degrees relative to the main panel **48** so as to extend into the water station **10**

when the door **38** is in the closed position. The mounting bracket **56** is provided with apertures **58** that correspond with bores defined in bosses **60** or other features of the filter hub **28** and through which threaded fasteners **62** are inserted to engage within the bores and fixedly secure the filter hub **28** directly to the mounting bracket **56** and, therefore, the door **38**. Accordingly, the filter **24** is indirectly connected to the mounting bracket **56**, but is supported thereby.

(18) At the distal end **42** of the door **38**, the lateral flanges **50** are further formed with inwardly extending flange portions that define support shelves **64**. The body **27** of the filter **24** may be received and supported on the support shelves **64** to reduce stress imparted to the threaded couplings **26**, **30** when the door **38** is in the closed position. A damping element **66**, such as a foam rubber or cork strip, may optionally be provided on the support shelves **64** to cushion the support of the filter **24** and minimize and vibrational noise that might otherwise result via contact between the body of the filter and the support shelves material support shelves **64**.

(19) The above description is meant to be illustrative of at least one preferred implementation incorporating the principles of the invention. One skilled in the art will really appreciate that the invention is susceptible to modification, variation and change without departing from the true spirit and fair scope of the invention, as defined in the claims that follow. The terminology used herein is therefore intended to be understood in the nature of words of description and not words of limitation.

Claims

1. A water providing station configured to be mounted to a wall and comprising: a housing; a basin directly supported by the housing and being located immediately above the housing; a water dispensing mechanism associated with the basin; a water supply line located within the housing; a filter coupled to the water supply line; a door mounted to a portion of the housing, the door being moveable between open and closed positions and extending downwardly from the housing in the open position and being flush with the housing in the closed position; and the filter being mounted to the door and supported by the door for movement with the door between the open and closed positions while remaining coupled to the water supply line.
2. The water providing station according to claim 1, wherein the door is pivotally mounted to the housing.
3. The water providing station according to claim 1, wherein one end of the door is pivotally engaged with the housing and at another portion of the door is releasably engaged with the housing.
4. A water providing station comprising: a housing; a basin supported by the housing; a water dispensing mechanism associated with the basin; a water supply line located within the housing; a filter coupled to the water supply line; a door mounted to a portion of the housing, the door being moveable between open and closed positions and extending downwardly from the housing in the open position and being flush with the housing in the closed position; and a filter hub supported by the door and being coupled to the water supply line in both the open and closed positions of the door, the filter hub being engaged with the filter and defining a fluid connection between the filter and the water supply line.
5. The water providing station according to claim 4, wherein the filter hub is fixedly fastened to the door for movement with the door between the open and closed positions.
6. The water providing station according to claim 4, wherein the filter hub is in threaded engagement with the filter.
7. The water providing station according to claim 1, wherein the door is pivotally engaged adjacent to one end of the door.
8. The water providing station according to claim 7, wherein the door includes a main panel defining an exterior surface, the exterior surface corresponding with an exterior surface of the housing in the closed position.

9. The water providing station according to claim 8, wherein the exterior surface is flush with the exterior surface of the housing in the closed position.
 10. The water providing station according to claim 8, wherein the door includes a mounting bracket, the filter being connected to the mounting bracket.
 11. The water providing station according to claim 10, wherein the filter is indirectly connected to the mounting bracket.
 12. The water providing station according to claim 10, wherein the mounting bracket extends away from the main panel on a side of the main panel opposite from the exterior surface of the main panel.
 13. The water providing station according to claim 10, further comprising a filter hub, the filter hub being mounted to the mounting bracket.
 14. The water providing station according to claim 13, wherein the filter hub is connected to the filter and indirectly connects the filter to the mounting bracket.
 15. The water providing station according to claim 1, wherein the door includes lateral flanges adjacent to at least one end of the door, the door being mounted to the housing by pivot pins extending through the lateral flanges.
 16. The water providing station according to claim 1, wherein the filter includes coupling port and a body portion, the body portion being supported by the door.
 17. The water providing station according to claim 16, wherein the door includes a main panel having support shelves provided along at least portions of lateral sides of the main panel.
 18. The water providing station according to claim 17, wherein the filter engages the support shelves.
 19. The water providing station according to claim 17, wherein the body of the filter engages the support shelves.
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